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Review article

Breakthroughs in AI and multi-omics for cancer drug discovery: A review

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ABSTRACT

Cancer is one of the biggest medical challenges we face today. It is characterized by abnormal, uncontrolled growth of cells that can spread to different parts of the body. Cancer is extremely complex, with genetic variations and the ability to adapt and evolve. This means we must continuously pursue innovative approaches to developing new cancer drugs. While traditional drug discovery methods have led to important breakthroughs, they also have significant limitations that make it difficult to efficiently create new, cost-effective cancer therapies. Integrating computational tools into the cancer drug discovery process is a major step forward. By harnessing computing power, we can overcome some of the inherent barriers of traditional methods. This review examines the range of computational techniques now being used, such as molecular docking, QSAR models, virtual screening, and pharmacophore modeling. It looks at recent advances in areas like machine learning and molecular simulations. The review also discusses the current challenges with these technologies and envisions future directions, underscoring how transformative these computational tools can be for creating targeted, new cancer treatments.

1. Introduction

Cancer remains one of the most formidable health challenges worldwide, characterized by uncontrolled cell proliferation and the potential to invade or spread to other parts of the body [1]. It encompasses over 100 different diseases, each with its unique molecular and cellular characteristics [2]. The global burden of cancer is substantial, with millions of new cases and deaths reported annually. According to the World Health Organization (WHO), cancer is the second leading cause of death globally, emphasizing the critical need for effective therapeutic interventions [3]. Anticancer drug discovery is of paramount importance in combating this pervasive disease [4]. The development of novel therapeutics not only aims to improve survival rates but also to enhance the quality of life for patients by minimizing side effects and overcoming drug resistance. The complexity of cancer, with its genetic heterogeneity and adaptive mechanisms (Fig. 1), necessitates a continuous and innovative approach to drug discovery. This pursuit involves identifying new drug targets, understanding the molecular mechanisms of action, and optimizing pharmacokinetics and pharmacodynamics to ensure efficacy and safety.

Traditional drug discovery methods, while having yielded significant breakthroughs, face numerous challenges that hinder the efficient and

cost-effective development of new anticancer therapies [5]. These challenges can be broadly categorized into scientific, technical, and financial aspects. Scientifically, the complexity of cancer biology, characterized by its high degree of genetic and phenotypic heterogeneity, often leads to a high attrition rate in drug development [6]. Identifying suitable drug targets is a crucial and challenging step that relies heavily on biological assays and experimental validation, which can be time-consuming and prone to errors [7]. Furthermore, cancer cells develop resistance to therapies through various mechanisms, such as genetic mutations, epigenetic changes, and alterations in drug metabolism, making it difficult to maintain long-term efficacy [8].

High-throughput screening (HTS) facilitates screening of several thousands of compounds but generates massive false positive/negative and is resource-intensive, time-consuming, and equipment-intensive [9]. It is a challenging process to fine-tune potential leads to achieve maximal activity and reduce toxicity; this often requires repeated cycles of synthesizing and assaying, which is time-consuming and expensive [10]. Many of the traditional preclinical models such as cell lines and animal models often do not accurately replicate human responses, which has been blamed for the high mortality rate of candidates in clinical trials [11]. From a financial aspect, the traditional approach to drug discovery has been criticized for being both costly and time-consuming;

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it can take over a decade to bring a drug from discovery to market, and the cost can reach billions of dollars [12]. The cost is involved from the research to the clinical trial phase and due to the high failure rate in drug development, there are significant losses. This risk is especially apparent in oncology due to the unique pathophysiology of cancer-related diseases.

The integration of computer-based programs into the process of developing anticancer drugs is a major improvement and helps to get over the drawbacks of lengthy conventional techniques [13]. Computer-aided drug design (CADD) [14–16] utilizes several computational methods as a tool for aiding the drug discovery process, including target identification, validation, lead identification, optimization and preclinical evaluation [17]. These tools are for instance Molecular docking; it is a computational approximation technique that predicts the preferred orientation of a drug candidate to its target receptor [18]. Furthermore, virtual screening makes it possible to screen large databases of compounds to generate leads quickly, while pharmacophore modeling helps to define crucial features of the molecules required for the biological activity [19].

Machine learning (ML) and artificial intelligence (AI) take these capabilities to another level by processing enormous amounts of data to identify patterns and make precise predictions [20]. These advanced algorithms can predict drug efficacy, identify possible side effects, and optimize drug candidates at a much faster and accurate way than other conventional approaches [21]. MD simulations offer detailed information at the atomic level regarding the dynamic properties of the drug-receptor interactions, enhancing the knowledge of binding and binding stability [22]. Computer-aided ADMET (absorption, distribution, metabolism, excretion, and toxicity) prediction tools enable determination of the pharmacokinetic and toxic effects of drug candidates prior to the later stages of drug development [23].

The purpose of this review is to provide a comprehensive overview of the various computer-aided tools employed in anticancer drug discovery and to highlight their applications, advantages, and limitations. By examining the integration of these tools into the drug discovery pipeline, this review aims to illustrate how they contribute to more efficient and effective development of anticancer therapies. The scope of the review encompasses the historical evolution of these tools, key methodologies such as molecular docking, QSAR, virtual screening, and pharmacophore modeling, as well as the latest advancements in machine learning, AI, and molecular dynamics simulations. Furthermore, the review will address the current challenges faced by these technologies, including their limitations and the future directions for their development and application in anticancer research. Through this detailed exploration, the review seeks to underscore the transformative impact of computer-aided tools on the field of anticancer drug discovery and their

potential to revolutionize the development of new, targeted therapies.

1.1. Historical background

It is therefore astonishing to realize that the discovery of anticancer drugs has been through certain steps and achievements that have brought us to where we are today. Chemotherapy started at the beginning of the twentieth century when chemotherapeutic agents like nitrogen mustard and antifolates were found [24]. These drugs being highly effective in killing rapidly dividing cells were not selective and led to many side effects [25]. The later part of the 20th century then saw the adoption of more selective treatments including hormone therapies for breast and prostate cancer that targeted specific biochemical processes used by the cancer cells to grow [26,27]. Molecular biology and genomics in the late Twentieth and early Twenty-first century changed the ways of identifying anticancer drugs [28]. Knowledge of oncogenes and tumor suppressor genes paved way for new drug targets and the development of targeted therapies like tyrosine kinase inhibitors and monoclonal antibodies [29]. They affirmed the necessity of a more reasonable approach to drug design that could meet the growing complexity of oncological pathology.

The use of computational methods in drug discovery can be traced back to 1960s and 1970s with the development of QSAR models [30]. These models used statistical methods which made it possible for one to predict the biological activity of a compound based on the chemical structure of the compound. Molecular docking which was introduced in the 1980s was another advancement in computational drug discovery because it made it possible to predict the preferred geometry or orientation of a given drug candidate when complexed to its target receptor [31]. This technique assisted in identifying lead compounds by determining the binding affinity as well as the stability of the complexes formed between drugs and their respective receptors.

It was only in the 1990s and the early years of the current millennium that the enhanced capability of computers and the availability of large biological databases paved way for the development of new and better computational techniques [32]. High-throughput virtual screening enabled large databases of compounds to be screened, and shortened the lead identification phase [33]. The integration of bioinformatics and cheminformatics contributed to the elucidation of the molecular architecture of biological systems and identification of new therapeutic targets and diagnostic biomarkers [34].

Recent advances in the field of machine learning and artificial intelligence or AI in the last decade have added on to this notion [35]. These technologies can sort through terabytes of data in record time and accurately detect relationships and likely scenarios. Using machine learning algorithms, drug efficacy is predicted, chemical structures are

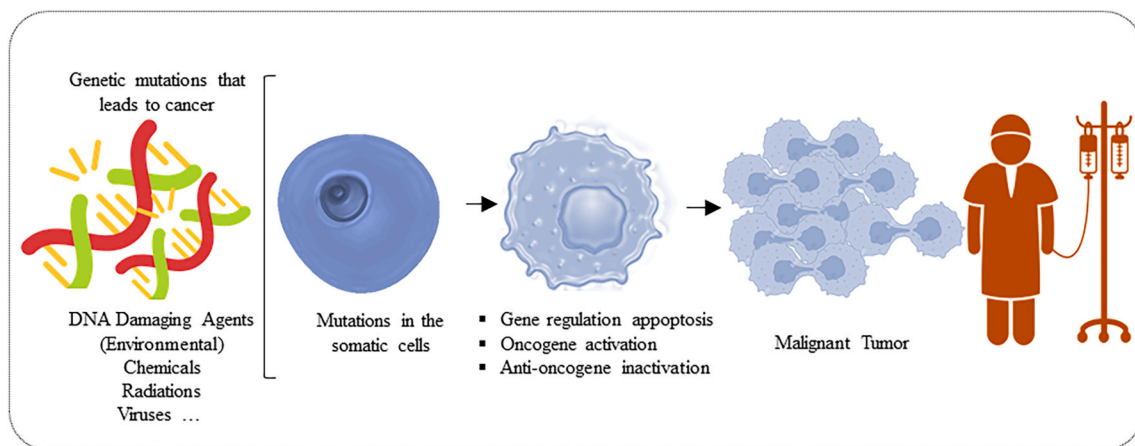


Fig. 1. Pathways to cancer development. The progression begins with damage to DNA, either through inherited genetic defects or environmental exposures, leading to disruptions in normal cellular functions.

optimized, and potential side effects are recognized, which improves the overall performance and success of drug discovery experiments [36]. In summary, historical timeline of anticancer drug discovery is characterized by ongoing process of development arising from scientific progression and technological advancement. The foundation for the contemporary highly advanced computational strategies was set by the early adopted models, which are crucial in managing the challenges of cancer and the creation of new and improved therapies.

1.2. Milestones in computer-aided anticancer research

These milestones helped in the development and progress of anticancer medications and their identification (Table 1). For instance, molecular docking played a significant role in discovery of Imatinib (Gleevec), a tyrosine kinase inhibitor in the treatment of CML [37]. Knowing how Imatinib binds to the BCR-ABL fusion protein allowed researchers to design a highly specific and potent drug that completely changed the treatment of CML. High-throughput virtual screening has also paid off; for example, Venetoclax (ABT-199) – the BCL-2 inhibitor for the CLL treatment (Fig. 2) [38]. Virtual screening enabled the researchers to screen thousands of compounds and come across Venetoclax that selectively bound to the BCL-2 protein which helps the cancer cells to survive. This combination of bioinformatics and cheminformatics led to the creation of Olaparib (Lynparza), a PARP inhibitor for the treatment of BRCA-mutated ovarian and breast cancers [39]. Scientists discovered that PARP played a crucial role for cancer cells with BRCA mutation and developed Olaparib to take advantage of it based on big data analysis on genetic information.

Further, HBCVTr, an end-to-end transformer with a deep neural network hybrid model for anti-HBV and HCV activity predictor from SMILES [40]. Molecular dynamics simulations have greatly led to the understanding of the various behaviors of drug-receptor interaction. One of major example is in designing Nivolumab (Opdivo), a PD-1 immune checkpoint inhibitor [41]. Molecular dynamics aided in revealing the changes of conformation and binding processes of Nivolumab so that its therapeutic potential could be enhanced. These particular studies demonstrate the potential of using computer-aided tools in the fight against cancer. Starting from the first QSAR models and moving into the AI and MD methods, every transition has led to the improvement of the efficiency, accuracy, and cost of the development of new anticancer drugs. The constant advancements of these tools also do guarantee to help in improving the fight against cancer and patient survival.

2. Computer-aided drug design (CADD)

Computer-aided drug design, also known as computer-aided molecular design allows the assessment of large numbers of chemicals, to estimate the binding between the drug candidates and targets and to predict the pharmacokinetic and toxic profiles of compounds in the early stages of the development cycle [42,43]. As shown in Fig. 3, CADD plays several important roles in the discovery of anticancer agents. Firstly, CADD increases the speed at which the desired compounds could be selected from large databases or libraries through computational high throughput screening as compared to experimental approaches [44]. Secondly, due to CADD, it is possible to describe some molecular mechanisms of the drug action and resistance and create new substances with increased activity and low toxicity [45]. Thirdly, since CADD involves molecular docking, QSAR modeling and pharmacophore modeling, it becomes easy to optimize lead compounds with high efficiency [46]. To sum up, CADD can improve the drug discovery process, increase the likelihood of the development, and help find and develop novel anticancer agents.

One successful example of computer-aided drug design for multi-cancer treatment is the development of Imatinib, which targets the BCR-ABL tyrosine kinase for treating chronic myeloid leukemia and gastrointestinal stromal tumors [47]. Erlotinib (Tarceva) inhibits the

Table 1

Key milestones in computer-aided anticancer research and their impact on drug discovery.

Year	Milestone	Description	Impact
1960s	Development of QSAR Models	Statistical correlation of chemical structures with biological activity.	Laid the groundwork for predictive modeling in drug discovery.
1970s	Introduction of Computer-Aided Drug Design (CADD)	Use of computers to design and optimize drug candidates.	Pioneered the use of computational techniques in drug development.
1980	Development of Molecular Mechanics	Use of physics-based models to predict molecular structures and properties.	Enhanced the accuracy of molecular modeling and simulations.
1981	Advent of Molecular Docking	Prediction of small molecule binding to target protein active sites.	Enhanced the lead identification process by providing insights into binding affinities and interactions.
1985	Introduction of Homology Modeling	Use of known protein structures to predict unknown structures.	Enabled the modeling of protein targets with limited structural information.
1990	High-Throughput Screening (HTS)	Automation of compound screening processes.	Increased the efficiency of lead compound identification.
1991	Development of Molecular Dynamics (MD) Simulations	Simulation of molecular interactions over time.	Provided dynamic insights into drug-receptor interactions.
1995	Introduction of Virtual Screening	Computational screening of large compound libraries.	Rapidly identified potential drug candidates.
1997	Development of Combinatorial Chemistry	Creation of large libraries of diverse molecules for screening.	Expanded the chemical space available for drug discovery.
1998	First Successful Use of Structure-Based Drug Design (SBDD)	Design of drugs based on the 3D structure of target proteins.	Led to the development of highly specific and effective drugs.
2000	Integration of Bioinformatics	Application of computational tools to biological data.	Facilitated the discovery of new drug targets and biomarkers.
2001	Launch of the Protein Data Bank (PDB)	Public repository of 3D structural data of proteins and nucleic acids.	Provided a valuable resource for structural biology and drug design.
2003	Completion of the Human Genome Project	Comprehensive map of the human genome.	Accelerated the identification of genetic mutations associated with cancer.
2005	Development of Pharmacophore Modeling	Identification of molecular features essential for biological activity.	Improved the identification and optimization of lead compounds.
2008	Introduction of Fragment-Based Drug Design (FBDD)	Use of small chemical fragments to develop potent drugs.	Enhanced the efficiency of lead discovery.
2010	Advancements in Machine Learning and AI	Application of ML and AI in predicting drug efficacy, toxicity, and designing drug molecules.	Improved the accuracy and efficiency of the drug discovery process.
2012	Development of Next-Generation Sequencing (NGS)	High-throughput sequencing technologies.	Enabled the rapid sequencing of cancer genomes for personalized medicine.
2014	CRISPR-Cas9 Genome Editing	Precise editing of genetic material.	Revolutionized functional genomics and target validation in cancer research.

(continued on next page)

Table 1 (continued)

Year	Milestone	Description	Impact
2015	Use of Deep Learning in Drug Discovery	Application of deep neural networks to predict drug interactions.	Enhanced the predictive power of computational models.
2016	AlphaFold by DeepMind	AI-driven prediction of protein structures.	Revolutionized the understanding of protein folding and structure prediction.
2017	Development of In Silico ADMET Prediction Tools	Computational prediction of absorption, distribution, metabolism, excretion, and toxicity properties.	Reduced the likelihood of late-stage drug development failures.
2018	Integration of Multi-Omics Data	Use of genomics, proteomics, and metabolomics in drug discovery.	Provided a comprehensive view of cancer biology for better target identification.
2019	AI-Based Drug Repurposing Platforms	Use of AI to find new uses for existing drugs.	Accelerated the discovery of anticancer properties in known compounds.
2020	AI-Driven Molecular Dynamics Simulations	Enhanced simulation techniques using AI to predict molecular interactions.	Improved the accuracy and speed of simulations.
2021	Use of Quantum Computing in Drug Discovery	Application of quantum algorithms to solve complex molecular problems.	Opened new possibilities for exploring chemical space.
2022	AI-Enhanced Structural Biology	Integration of AI with experimental structural techniques like cryo-EM.	Provided more accurate and detailed structural data for drug design.

epidermal growth factor receptor (EGFR), beneficial in non-small cell lung cancer and pancreatic cancer [48]. Crizotinib (Xalkori) targets ALK and ROS1 receptors in non-small cell lung cancer [49]. Palbociclib (Ibrance) inhibits cyclin-dependent kinases CDK4 and CDK6, crucial in hormone receptor-positive breast cancer [50]. Alectinib (Alecensa) targets ALK and is used for lung cancer [51]. Vemurafenib (Zelboraf) targets BRAF mutations in melanoma [52]. Trastuzumab (Herceptin), designed to target HER2, treats breast cancer [53]. Bevacizumab (Avastin) inhibits VEGF pathways to treat various solid tumors [54]. Lastly, Pembrolizumab (Keytruda) targets PD-1, a checkpoint inhibitor effective against melanoma, lung cancer, and head and neck cancers [55]. These examples underscore how CADD has revolutionized oncology by allowing for targeted therapies that improve treatment outcomes and reduce side effects compared to conventional

chemotherapies.

2.1. Structure-based drug design (SBDD) with machine learning enhancements

Structure-Based Drug Design (SBDD) is one of the most efficient methods of drug discovery that uses structural information of the targets such as proteins to optimize drugs candidates [56]. Therefore, by incorporating more structural information, chemists are able to observe specific binding sites and molecular recognition between the drug and the target yielding improved and selective drug molecules [57]. Some of the phases for SBDD include target identification and validation, structural determination by X-ray crystallography, Nuclear Magnetic Resonance spectroscopy, cryo-electron microscopy, binding site analysis, molecular docking, and the lead optimization phase [58]. This method enables a more rational plan to be taken in drug discovery, which enhances the chances of success by embracing molecular identification. Therefore, SBDD has several advantages over other methods of discovering anticancer drugs. They can produce highly specific reagents that are more likely to have minimum interferences hence better treatments [59]. It also improves the drug discovery process by using mathematical models that quickly identify and adjust possible drug targets rather than using experiments. SBDD enables the modification of drug molecules depending on their structures, which can enhance the effectiveness of drugs and minimize their toxicity.

Recently, Cui et al. [60] performed SBDD of crizotinib (PF-02341066), a potent and selective dual inhibitor of mesenchymal-epithelial transition factor (c-MET) kinase. Similarly, Gao et al. [61] proposed peptide deformylase inhibitors against cancer through SBDD. Yadav et al. [62] reported that paclitaxel, etoposide and topotecan as anticancer molecules after performing SBDD and molecular docking studies Crizotinib (Xalkori) is an ALK inhibitor for NSCLC patients with ALK gene rearrangements and was identified using the structure of the ALK kinase domain [63]. This structure-guided approach led to the development of a drug that acts on ALK-positive NSCLC and opens a new therapeutic avenue for patients with this genetic alteration. In conclusion, SBDD plays an essential role in the discovery of new anticancer drugs, which enables the development of targeted drugs in a rational manner [64]. With detailed structural information of biological targets, SBDD is possible to design and optimize drug candidates with high selectivity and affinity, which ultimately result in better therapeutic outcomes. A few examples of the successful application of SBDD in oncology are Imatinib, Vemurafenib, Nivolumab, Crizotinib, and Olaparib, which give promises for better future of cancer therapies.

In a recent study, researchers have used the SBDD integrated with machine learning to accelerate the lead optimization of an enzyme

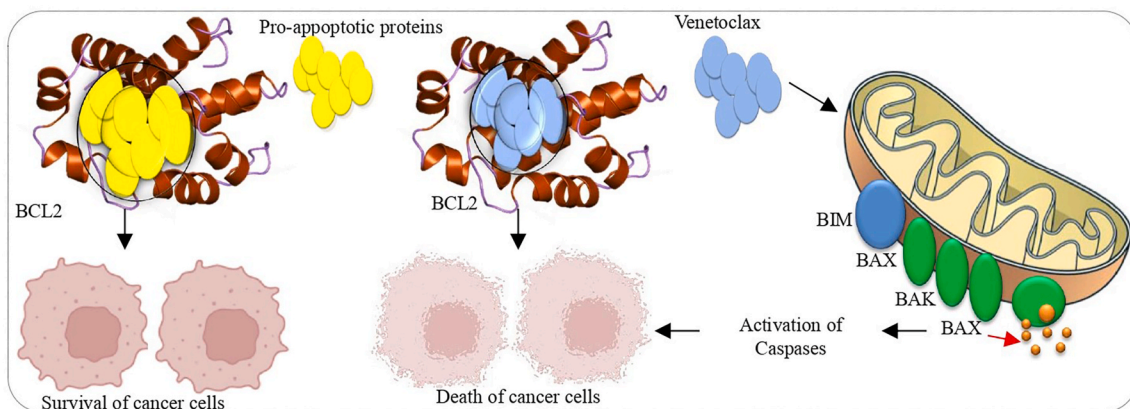


Fig. 2. Molecular Mechanism of Venetoclax in Inducing Apoptosis in Cancer Cells. Venetoclax, represented in blue, specifically targets and binds to BCL2, altering its function.

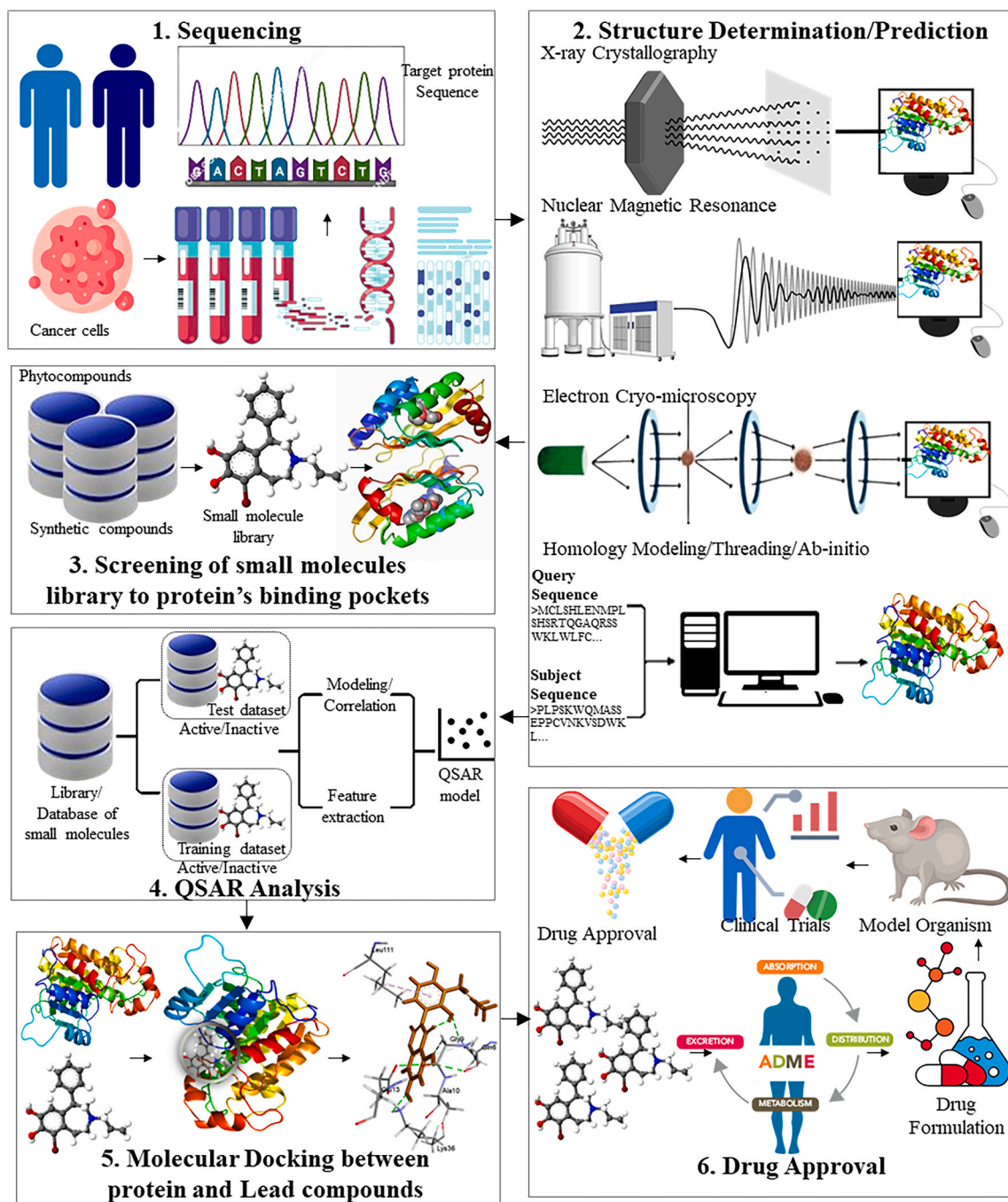


Fig. 3. Integrated Workflow for Drug Discovery and Design. It outlines a streamlined workflow in the drug discovery process, starting from the identification of target proteins and genes related to diseases such as cancer.

inhibitor for a vital protein associated with cancer development [65]. Using the structural information of the protein, the team was able to determine important binding zones and estimate the binding constants of the compounds. The ML models were trained on existing datasets, which enabled predicting how any changes in the molecular structure would affect its interaction with the target. This approach therefore substantially saved time and effort in terms of lead optimization and delivered a hopeful candidate that advanced to preclinical trials faster than was possible from a traditional approach.

2.2. Ligand-based drug design (LBDD) leveraging machine learning techniques

Ligand-based drug design (LBDD) is an approach to drug discovery that seeks to design and optimize drug candidates based on information concerning other molecules (ligands) that interact with the target protein [66]. Unlike SBDD, LBDD does not require the initial information about the structure of the target protein. Instead, it bases its arguments on chemical and biological information of the known ligands to estimate the nature and behavior of other similar molecules [66]. It is particularly useful where the structure of the target protein is unknown or is not easily determined. The methods applied in LBDD include QSAR modeling, pharmacophore modeling, and machine learning-based

screening. In QSAR modeling, chemical structures are examined statistically to analyze the relationship between molecular descriptors and biological activities of compounds [67]. This helps the researchers to be in a position to foretell how new compounds are going to behave depending on their structures. Pharmacophore modeling defines the features in a molecule needed for it to interact with a specific biological entity and helpful in the design of new chemical entities containing the required pharmacophoric group to elicit a desired biological response. Virtual screening involves using computational methods (Table 2) to evaluate the libraries of compounds against the target protein to generate a list of good drug candidates rapidly.

For instance, Thalidomide, initially used for morning sickness, was repurposed for treating multiple myeloma based on its anti-angiogenic and immunomodulatory properties [68]. This repurposing was driven by understanding its ligand interactions with the cereblon protein, affecting various signaling pathways crucial for myeloma cell survival. Another example is the development of Bicalutamide, a non-steroidal anti-androgen used for prostate cancer. It was designed by studying the molecular structures of testosterone and other androgens that bind to the androgen receptor, allowing for the synthesis of an effective antagonist [69]. Another estrogen agonist that was derived from modifying the ligands of estrogen receptors is raloxifene, which is used in preventing breast cancer [70]. Flutamide, another anti-androgen used in prostate cancer treatment [71], was developed through knowledge of how androgens attach themselves to their receptors and the creation of a molecule that would block this action. Finally, the therapeutic application of Aromatase inhibitors such as Anastrozole includes blocking the activity of the aromatase enzyme [72], which plays a crucial role in estrogen biosynthesis. Through studying how the enzyme worked with natural substrates, potent inhibitors that lowered estrogen levels were created, useful in managing ER + breast cancer. These examples highlight how ligand-based approaches can refine drug properties to target specific cancer pathways effectively. Raltegravir as a member of the antiretroviral class of drugs shows how LBDD can evolve enzyme inhibitors [73]. This is true because scientists used standard integrase inhibitors and pharmacophore templates to synthesize Raltegravir, an effective HIV integrase inhibitor.

Sorafenib is an orally active multi-kinase inhibitor that is effective in the treatment of hepatocellular carcinoma [74]. Sorafenib was developed using LBDD, in particular, pharmacophore modeling and QSAR, for the purpose of creating a compound which would target multiple kinases involved in cancer cell proliferation and formation of new blood vessels. This led to the synthesis of Sorafenib which acts on the RAF kinase, VEGFR and the PDGFR that endowed the compound with wide spectrum antitumor activity [75]. In this regard, based on the available chemical and biological information of known ligands, LBDD allows for rapid identification and enhancement of efficacious anticancer agents. Some of the drugs that have been developed using LBDD include venetoclax, olaparib, raltegravir, sorafenib, and gefitinib, which demonstrates the efficacy of LBDD in designing therapies for patients. Thus, LBDD will remain an essential tool for the search for new and effective cancer treatments as computational methods develop.

3. Quantitative structure-activity relationship (QSAR)

QSAR is also referred to as quantitative structure-activity relationship or quantitative structure-property relationship and it is a computational method that predicts the activity of compounds based on their structure employing statistical models (Table 3) [67]. The basic principle of QSAR is the fact that the chemical structure of a compound determines its activity. QSAR models can be used for the evaluation of the new untested compound activity from the SAR of the series of compounds [106]. QSAR is very important in the discovery of anticancer drugs because of its efficiency in the identification of leads. In addition, QSAR can predict possible off-target activity and ADMET profile of the compounds under analysis so that the further steps of drug development

are accelerated [107].

2D QSAR models make use of representations of the molecule in two dimensions in terms of molecular weight, hydrophobicity, charge density, and any other specific functional groups [108]. These models are less sophisticated, and less computationally demanding, which makes them suitable for the first cut of the analysis. 3D QSAR models require the consideration of position of the atom in the molecule [109]. In 3D QSAR, methods like Comparative Molecular Field Analysis (CoMFA) and Comparative Molecular Similarity Indices Analysis (CoMSIA) provide much enhanced information of the molecular fields, which may be valuable for understanding the binding affinity and selectivity of the drug molecules [110].

4D QSAR is an advancement on 3D QSAR which incorporates conformational flexibility and movement over time of molecules into consideration hence offering better insight of how molecules behave under dynamic biological conditions [111]. In addition to this, 5D QSAR takes into consideration solvation states because solvents alter interaction and stability of molecules and 6D QSAR involves different protonation states of molecules because the change in charge and structure due to change in pH level is very crucial in determining the activity [112].

Additionally, QSAR studies were integral in designing more effective derivatives of Camptothecin, a topoisomerase inhibitor, by identifying structural features that maximize the drug's stability and interaction with the topoisomerase enzyme [133,134]. In the realm of anti-metabolites like Methotrexate for treating various cancers [135], QSAR models were employed to examine the relationship between chemical structure variations and their inhibitory activity against the dihydrofolate reductase enzyme, enhancing drug design efficiency. Another example is the discovery of HIV protease inhibitors like Saquinavir where researchers applied 3D QSAR models to study the interactions of different inhibitors with the HIV protease enzyme [136]. This analysis was useful in identifying the structural aspects necessary for good inhibition, which helped in the development of Saquinavir with improved affinity and selectivity. In the studies concerning the aromatase inhibitors applied in the treatment of estrogen receptor-positive breast cancer, the 3D QSAR methods including the CoMFA and CoMSIA were used for the determination of the molecular characteristics essential for the effective aromatase enzyme inhibition that led to the creation of the drugs like Anastrozole. QSAR models have also been applied in the prediction of toxicity of environmental chemicals. For example, researchers used 2D QSAR models to predict the carcinogenicity of different compounds and the structural alerts that are relevant to it; this made regulatory agencies screen and prioritize chemicals based on the results for further testing and regulation. Thus, QSAR models are quite valuable in drug discovery as they allow for building a mathematical model that can predict the activity of new compounds based on a structure activity relationship. The use of 2D, 3D, 4D, 5D, and 6D QSAR models gives researchers further understanding of molecular interactions, refinement of drug lead, and assessment of biological activity. The use of QSAR in the development of other drugs such as Imatinib, Saquinavir, and Anastrozole makes it relevant in anticancer drug discovery and other fields. Therefore, QSAR will persist as a key tool in the field of computational methods and technologies for drug discovery of new and effective treatments.

3.1. Machine learning techniques in QSAR modeling

QSAR modeling has been improved by using the ML approaches for the prediction of biological activity from chemical structure [137]. These techniques use big data and complex algorithms, which may help to find the connections, that could be missed when using traditional statistical methods [138]. The simplest of them is linear regression, which describes the dependence of one variable from one or more other variables, and it often used for the basic QSAR analysis. Random forest, an example of ensemble learning approach, builds many decision trees at the learning phase and provides the classification mode (in case of

Table 2

Tools used for ligand-based drug designing.

Category	Tool Name	Type	Primary Use	Features	Platform	Developer	License	Reference
Structure-Based	AutoDock	Molecular Docking	Predicting binding modes	Flexible docking, grid-based	Windows, Mac, Linux	The Scripps Research	Open Source	[76]
Structure-Based	Glide	Molecular Docking	Predicting binding affinities	High-throughput docking	Windows, Linux	Schrödinger LLC	Commercial	[77]
Structure-Based/ Ligand-Based	MOE (Molecular Operating Environment)	Integrated Suite	Drug discovery & design	Docking, homology modeling	Windows, Mac, Linux	Chemical Computing Group	Commercial	[78]
Structure-Based	GOLD	Molecular Docking	Predicting ligand binding	Genetic algorithm-based docking	Windows, Linux	Cambridge Crystallographic Data Centre	Commercial	[79]
Structure-Based	Rosetta	Protein Modeling	Protein structure prediction	Homology modeling, docking	Linux	Rosetta Commons	Open Source	[80]
Structure-Based	Chimera	Visualization	Molecular graphics	Visualization, analysis	Windows, Mac, Linux	UCSF	Open Source	[81]
Structure-Based	PyMOL	Visualization	Molecular visualization	3D rendering, scripting	Windows, Mac, Linux	Schrödinger LLC	Open Source	[82]
Structure-Based	Maestro	Visualization	Molecular modeling	Integrated suite	Windows, Mac, Linux	Schrödinger LLC	Commercial	[83]
Structure-Based	GROMACS	Molecular Dynamics	Simulation of biomolecules	High-performance MD	Windows, Mac, Linux	GROMACS development teams	Open Source	[84]
Structure-Based	CHARMM	Molecular Dynamics	Simulation of biomolecules	Comprehensive force fields	Windows, Mac, Linux	Harvard University	Open Source	[85]
Structure-Based	AMBER	Molecular Dynamics	Simulation of biomolecules	Extensive force fields	Windows, Mac, Linux	University of California, San Francisco	Commercial	[86]
Structure-Based	Phenix	X-ray Crystallography	Protein structure determination	Automated structure solution	Windows, Mac, Linux	Lawrence Berkeley National Laboratory	Open Source	[87]
Structure-Based	CCP4	X-ray Crystallography	Protein structure determination	Comprehensive suite	Windows, Mac, Linux	Collaborative Computational Project Number 4	Open Source	[88]
Structure-Based	COOT	Model Building	Model building for X-ray data	Interactive model building	Windows, Mac, Linux	MRC Laboratory of Molecular Biology	Open Source	[89]
Structure-Based	MODELLER	Homology Modeling	Comparative protein modeling	Automated model generation	Windows, Mac, Linux	University of California, San Francisco	Open Source	[90]
Structure-Based	SWISS-MODEL	Homology Modeling	Comparative protein modeling	Web-based service	Web-based	Swiss Institute of Bioinformatics	Free	[91]
Structure-Based	I-TASSER	Protein Modeling	Protein structure prediction	Fold recognition, ab initio	Web-based	University of Michigan	Free	[92]
Ligand-Based	OpenEye OMEGA	Conformational Search	Generation of 3D conformations	Rapid conformer generation	Windows, Mac, Linux	OpenEye Scientific	Commercial	[93]
Ligand-Based	Forge	QSAR	3D QSAR modeling	Field-based QSAR	Windows, Linux	Cresset Group	Commercial	[94]
Ligand-Based	LigandScout	Pharmacophore Modeling	Pharmacophore-based virtual screening	Feature-based pharmacophores	Windows, Mac, Linux	Inte GmbH	Commercial	[95]
Ligand-Based	ROCS	Shape Comparison	Shape-based screening	Rapid shape comparison	Windows, Mac, Linux	OpenEye Scientific	Commercial	[96]
Ligand-Based	Phase	Pharmacophore Modeling	3D pharmacophore modeling	Hypothesis generation, screening	Windows, Linux	Schrödinger LLC	Commercial	[97]
Ligand-Based	Catalyst	Pharmacophore Modeling	Pharmacophore-based screening	Feature-based pharmacophores	Windows, Linux	Accelrys Inc.	Commercial	[98]
Ligand-Based	Sybyl	Integrated Suite	Drug discovery & design	QSAR, molecular modeling	Windows, Linux	Tripes Inc.	Commercial	[99]
Ligand-Based	ChemAxon JChem	Chemical Informatics	Chemical database management	Structure search, property prediction	Windows, Mac, Linux	ChemAxon	Commercial	[100]
Ligand-Based	KNIME	Workflow Management	Data analytics, integration	Modular data processing	Windows, Mac, Linux	KNIME AG	Open Source	[101]
Ligand-Based	SwissADME	ADME Prediction	Toxicity and metabolism prediction	QSAR-based models	Windows	Simulations Plus	Commercial	[102]
Ligand-Based	Discovery Studio	Integrated Suite	Drug discovery & design	QSAR, pharmacophore modeling	Windows	BIOVIA	Commercial	[103]
Ligand-Based	Schrodinger's QikProp	ADMET Prediction	Physicochemical property prediction	Extensive ADMET profiling	Windows, Mac, Linux	Schrödinger LLC	Commercial	[104]
Ligand-Based	Chemdraw	Chemical Drawing	Molecular structure drawing	Structure visualization	Windows, Mac	PerkinElmer	Commercial	[105]

Table 3

Tools used for QSAR in drug discovery.

QSAR Type	Tool Name	Developer	Primary Use	Key Features	Platform	License	Reference
2D QSAR	QSAR Toolbox	OECD (Organization for Economic Co-operation and Development)	QSAR modeling	Chemical structure analysis, endpoint predictions	Windows, Mac, Linux	Free	[113]
2D QSAR	Dragon	Talete SRL	Molecular descriptor calculation	Extensive library of 2D descriptors	Windows	Commercial	[114]
2D QSAR	PaDEL-Descriptor	National University of Singapore	Descriptor calculation	2D and 3D molecular descriptors	Windows, Mac, Linux	Open Source	[115]
3D QSAR	CoMFA	Tripos Inc.	3D QSAR modeling	Comparative Molecular Field Analysis	Windows, Linux	Commercial	[116]
3D QSAR	CoMSIA	Tripos Inc.	3D QSAR modeling	Comparative Molecular Similarity Indices Analysis	Windows, Linux	Commercial	[117]
3D QSAR	MOE	Chemical Computing Group	Integrated QSAR and modeling	3D QSAR, pharmacophore modeling	Windows, Mac, Linux	Commercial	[78]
3D QSAR	Forge	Cresset Group	3D QSAR modeling	Field-based QSAR modeling	Windows, Linux	Commercial	[94]
3D QSAR	PHASE	Schrödinger LLC	3D QSAR modeling	Pharmacophore modeling, 3D QSAR	Windows, Linux	Commercial	[97]
3D QSAR	Open3DQSAR	Open3DQSAR community	3D QSAR modeling	Open-source CoMFA/CoMSIA implementation	Windows, Mac, Linux	Open Source	[118]
4D QSAR	Quasar	University of Florida	4D QSAR modeling	Conformational flexibility, dynamic behavior	Windows, Mac, Linux	Commercial	[119]
4D QSAR	S4MPLE	University of Florida	4D QSAR modeling	Simulation of molecular populations	Windows, Mac, Linux	Commercial	[120]
4D QSAR	Chem4D	Various	4D QSAR modeling	Incorporates molecular dynamics	Windows, Mac, Linux	Varies	[121]
5D QSAR	GALAHAD	Tripos Inc.	5D QSAR modeling	Considers solvation states	Windows, Linux	Commercial	[122]
6D QSAR	Catalyst	BIOVIA (Accelrys Inc.)	6D QSAR modeling	Includes protonation states	Windows, Linux	Commercial	[98]
6D QSAR	GOLD Suite	Cambridge Crystallographic Data Centre (CCDC)	6D QSAR modeling	Extensive ligand flexibility	Windows, Linux	Commercial	[79]
All QSAR	KNIME Analytics Platform	KNIME AG	Data analytics, integration	Modular data processing, QSAR modeling	Windows, Mac, Linux	Open Source	[101]
All QSAR	ChemAxon JChem	ChemAxon	Chemical database management	Structure search, property prediction	Windows, Mac, Linux	Commercial	[123]
All QSAR	Pipeline Pilot	BIOVIA	Data analysis and modeling	Integration of diverse data sources	Windows	Commercial	[124]
All QSAR	OpenTox	OpenTox community	Toxicity prediction	Open-source QSAR models for toxicity	Windows, Mac, Linux	Open Source	[125]
All QSAR	Chem3D	PerkinElmer	Molecular modeling	3D structure visualization, QSAR	Windows, Mac	Commercial	[126]
All QSAR	Spartan	Wavefunction Inc.	Computational chemistry	QSAR, molecular modeling	Windows, Mac, Linux	Commercial	[127]
All QSAR	RStudio	RStudio Inc.	Statistical computing	Data analysis, QSAR modeling	Windows, Mac, Linux	Open Source	[128]
All QSAR	MATLAB	MathWorks	Numerical computing	Data analysis, QSAR modeling	Windows, Mac, Linux	Commercial	[129]
All QSAR	Simca	Sartorius Stedim Data Analytics	Multivariate data analysis	QSAR modeling, statistical analysis	Windows	Commercial	[130]
All QSAR	Chemaxon Reactor	ChemAxon	Chemical reaction enumeration	QSAR, reaction modeling	Windows, Mac, Linux	Commercial	[131]
All QSAR	WEKA	University of Waikato	Machine learning and data mining	QSAR, classification, regression	Windows, Mac, Linux	Open Source	[132]

classification problem) or mean response (in case of regression problem) of the trees [139]. They are good with large datasets and do not get easily overfitted. SVMs are employed in classification as well as regression analysis in order to identify the right hyperplane that will act as a separator of data classes. Efficient in the high dimensions, the SVM is particularly helpful when the number of features is larger than the number of samples [140].

Neural networks are a class of computational models that are based on biological neurons and consist of layers of nodes (neurons) capable of capturing non-linear dependencies. Neural networks with multiple layers, known as deep learning, have proven to be particularly effective in QSAR modeling. k-Nearest Neighbors (k-NN) assigns a data point to the class of its neighbors, and is easy to implement and efficient for small datasets, but less suitable for large ones due to computational complexity. GBM builds models iteratively, adding a new model to correct for the errors made by previous models, and has been applied to numerous prediction problems, including QSAR. Gaussian processes are

non-parametric models for regression that can be used for QSAR when it is necessary to make an assessment of the uncertainty in the given prediction [141,142].

The QSAR modeling for the prediction of the activity of anticancer compounds has been revolutionized with the help of machine learning techniques. For instance, the prediction of kinase inhibitors, which is crucial in cancer treatment, has been improved by the use of ML techniques. Simeon et al. [143] used random forest and SVM to build the QSAR models for a set of kinase inhibitors. These models helped in the understanding of the activity of new compounds and played an important role in the discovery of potent kinase inhibitors. Another compound with anticancer activity that was taken through ML techniques was the Histone deacetylase (HDAC) inhibitors. QSAR models for HDAC inhibitors were also developed by using deep learning methods by Li et al. [144]. By using the trained neural network model, they were able to predict the HDAC inhibition with high accuracy which will help identify other anticancer agents.

QSAR modeling has also been applied on natural products that have been found to contain a large number of anticancer compounds. Majid et al. [145] used k-NN and SVM for predicting the antiproliferative activity of natural compounds. All of their QSAR models highlighted several compounds for further study in the future. Further, the application of QSAR models based on ML brought about the identification of multitarget drugs, which are capable of interacting with multiple targets associated with cancer. For identifying compounds that can modulate multiple cancer-related targets, Planche et al. (2016) employed random forest and neural networks based QSAR models for multitarget anticancer drugs [146].

Additionally, the QSAR models of PI3K inhibitors have been established using deep learning for the PI3K/AKT/mTOR pathway which is overexpressed in cancer [147]. QSAR models for PI3K inhibitors were established by Yu et al. [148] and might help in predicting the activity of new inhibitors which can be useful in the design of new anticancer agents targeting this pathway. Thus, the range of the machine learning approach has expanded the possibilities of QSAR modeling and continues to be a valuable asset in the creation of anticancer drugs. Using big data and complex methods of analytics, the QSAR models based on ML can recognize the potency of compounds and support the creation of new drugs [149,150]. The usefulness of these models in identifying kinase inhibitors, HDAC inhibitors, tubulin polymerization inhibitors, and multitarget anticancer drugs supports the effectiveness of these models in identifying new cancer therapeutic agents. The improvement of the mentioned machine learning methods and their utilization in QSAR modeling can undoubtedly promote the improvement of anticancer drug discovery.

3.2. Virtual screening along with machine learning as new frontiers in cancer drug discovery

Virtual screening is a critical step in the discovery of cancer drugs since it can help find potential anticancer compounds within a short period and at a low cost [151]. For Possible drugs can be discovered with more efficacy and confidence when researchers integrate virtual screening with ML algorithms. They can process big data, recognize patterns, and estimate binding affinities with a better efficiency than traditional approaches. The combination of ML and virtual screening helps advance the drug discovery process further and more effectively, raising the prospects of identifying viable anticancer products. Techniques such as neural networks, support vector machines (SVM), random forest and deep learning methods are commonly used to enhance virtual screening. Such models can be trained on big databases of chemical compounds and their biological activity and can predict the properties of new compounds. For example, deep learning models can forecast the configuration of proteins and ligands to determine how any compound will engage with its target. The fact that the data can be managed in higher dimensions and, thus, the ability to model the interdependencies makes the ML useful for virtual screening.

Several reports have incorporated virtual screening with machine learning into the process of discovering anti-cancer drugs, which employ diverse approaches and targets. For instance, one of the recent works used virtual screening in conjunction with machine learning to identify novel inhibitors of the KRAS G12C mutation, which occurs in lung cancer [152,153]. Another study also applied deep learning models to improve the virtual screening outcomes for ligands targeting the PD-1/PD-L1 immune checkpoint that is vital in immunotherapy for melanoma [154]. Furthermore, structure-based virtual screening was integrated with machine learning to discover new PARP1 inhibitors for the treatment of ovarian and breast cancer [155]. In addition, Tondar et al. [156] proposed a convolutional neural network approach to categorize the results of virtual screening for BCL-2 inhibitors for enhanced apoptosis in tumor cells. Furthermore, a large-scale study applied virtual screening to assemble a collection of kinase inhibitors, which went through a supervised learning algorithm to identify novel

treatments for colorectal cancer.

Multi-target kinase inhibitors are compounds that are believed to inhibit multiple kinases involved in the development of cancer. Such inhibitors were reported by Ma et al. [157] in a study that used machine learning based virtual screening. Gradient boosting machines and other machine learning algorithms were employed, and different datasets with information on diverse kinase inhibitors were used to train these models. The models determined new compounds' binding affinities that were further validated through virtual screening. It is evidenced from this approach that ML can be used to discover complex drugs because it identified several potential multi-target kinase inhibitors.

Gefitinib, an EGFR inhibitor, is used in the treatment of non-small cell lung cancer. Researchers used machine learning techniques to improve virtual screening of EGFR inhibitors [158]. The ML algorithms used training data from structural and activity of known EGFR inhibitors to estimate the binding affinities of novel compounds with high accuracy. As a result of combining the use of ML and virtual screening, researchers were able to develop more effective EGFR inhibitors for treating NSCLC [159]. Virtual screening combined with machine learning has brought significant changes to cancer drug discovery. Thus, ML algorithms help researchers process large datasets, identify intricate patterns, and forecast binding affinities of compounds more effectively than existing techniques. This synergy shortens the time needed to discover promising compounds, lowers costs, and raises the probability of creating effective anticancer agents. Some examples where machine learning has been used in combination with virtual screening include the discovery of Venetoclax, PI3K inhibitors, PARP inhibitors, multi-target kinase inhibitors, and EGFR inhibitors.

4. Integrating machine learning for enhanced insights in molecular docking and cancer discovery

Molecular docking is a computational method that predicts the binding mode of a ligand to a receptor and the biological potency (binding energy) [31]. This method is important in drug discovery as it can predict and rank the binding of drug candidates to the target proteins associated with diseases like cancer [160]. Molecular docking comprises the conformational search of both the ligand and the receptor and a subsequent scoring step to estimate the binding energy. The binding energy (ΔG_{bind}) can be calculated using the equation:

$$\Delta G_{bind} = E_{complex} - (E_{receptor} + E_{ligand}) \quad (1)$$

where $E_{complex}$, $E_{receptor}$, and E_{ligand} are the energies of the complex, receptor, and ligand, respectively. The binding free energy (ΔG_{bind}) is related to the binding affinity (K_d) by the equation:

$$\Delta G_{bind} = -RT \ln K_d \quad (2)$$

where R is the gas constant and T is the temperature in Kelvin. Additionally, the scoring function used in molecular docking can be expressed as:

$$S = \sum_{i=1}^n w_i f_i \quad (3)$$

where S is the total score, w_i are weights, and f_i are various scoring terms that include van der Waals interactions, electrostatic interactions, and desolvation energies.

This technique not only helps in elucidating the molecular mechanisms of the drug action, but also facilitates the rational drug discovery process by ranking compounds for further experimental testing. Recent advancements in combining ML with molecular docking have improved the reliability and speed of drug discovery procedures [160]. The data obtained from the docking simulations can be processed by machine learning algorithms to discern patterns and improve the scoring functions used to model the binding affinities (Fig. 4). This collaboration makes it easier and faster to identify the right drug candidates given that

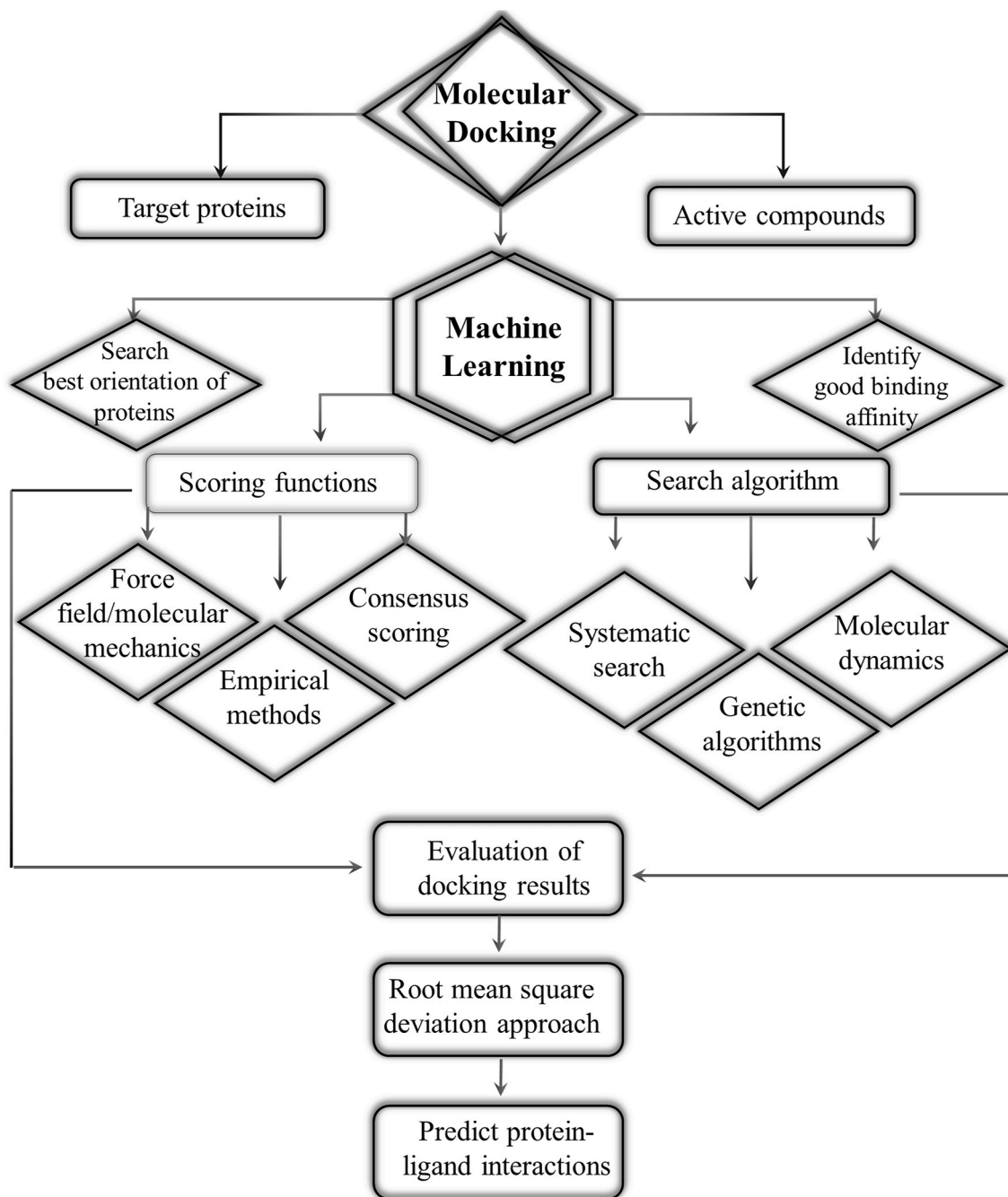


Fig. 4. Structured overview of the integration of molecular docking and machine learning in the discovery and development of pharmaceutical compounds.

the number of false positives is minimized in the process. For example, Das et al. [161] combined machine learning with docking simulations to identify potent inhibitors of the EGFR kinase, crucial in treating lung cancer. Samlimi et al. [158] utilized machine learning to optimize docking results for VEGF inhibitors, leading to advanced drug candidates for colorectal cancer. Similarly, research involving CDK6, a key regulator of cell cycle progression in cancer cells, utilized AI-enhanced docking to streamline the identification of novel inhibitors with potential applications in cancer [162]. A creative approach was seen in a study that applied reinforcement learning to improve docking strategies, targeting the androgen receptor in prostate cancer, showing significant improvements in binding affinity predictions.

In the realm of breast cancer, researchers developed a hybrid model integrating deep learning with docking data to predict the efficacy of ER α inhibitors, yielding promising leads that could outperform existing

therapies [163]. Another study focused on melanoma targeted the BRAF V600E mutation using a machine learning model to analyze and predict the best binding modes from thousands of docking simulations [164], facilitating faster lead optimization. In brain cancer, a significant study employed machine learning algorithms to refine docking outputs for MGMT, an enzyme involved in tumor resistance to certain chemotherapies, resulting in more accurate predictions of compound efficacy. Altogether, these studies have demonstrated how machine learning and molecular docking can increase the efficiency and the efficacy of the drug discovery procedures, as well as generating novel and effective therapeutic approaches to fight various types of cancers. The successful application of the integrated approach in discovering inhibitors for targets like AR, KRAS G12C, PI3K/AKT/mTOR and BCL-2 proves the role of the integrated approach in oncology.

5. Integration of multi-omics data with computational tools for anticancer drug discovery

The combination of multi-omics analysis with computational models in the development of anticancer drugs is a novel paradigm in cancer pharmacology. Multi-omics is the combination of several categories of omics which include genomics, transcriptomics, proteomics, metabolomics, and epigenomics [165]. With the help of computational approaches in working with omics data, it is possible to identify new targets for therapy and individualized approach in cancer therapy within the framework of multidimensional omics analysis. It is possible to identify cancer genes, proteins, metabolites, and even changes in epigenetic marks and thus better understand the development of cancer by using multi-omics data. For example, combining genomic and transcriptomic data can show specific genetic alterations and gene expression changes associated with cancer. Application of proteomics and metabolomics will assist in determining the proteins that are up-regulated or down-regulated in cancer cells and alterations in metabolic pathways respectively. By employing epigenomic data it is possible to identify changes in DNA methylation and histone modifications that control genes in cancer. To study the multi-omics data, various approaches like machine learning, network analysis, and pathways enrichment analysis are applied. These tools help to identify the basal levels of the genes which are involved in the development of the common molecular networks and signaling pathways in cancer cells, select the possible therapeutic targets, and estimate the probability of the favourable outcome after the specific course of treatment. Multi-omics integration and the use of computational methods therefore advances the identification of novel therapeutic targets and the development of targeted anticancer drugs (Fig. 5).

5.1. Case studies (discovery of BRD4 inhibitors)

Bromodomain and extra-terminal (BET) protein family member BRD4 is a newly identified target for cancer treatment. Scientists combined genomic, transcriptomic, and epigenomic datasets to nominate BRD4 as a central modulator of oncogenic transcriptional networks. To prioritize BRD4 as a therapeutic target, computational tools were

applied to analyze these datasets. Further, through virtual screening and molecular docking experiments efficient small molecule inhibitors of BRD4 were discovered and the drugs like JQ1 was found to be effective in preclinical cancer models [166].

5.2. Synthetic lethality screening

Synthetic lethality is defined as the condition under which the inactivation of two genes is lethal whereas the inactivation of only one of these genes is not lethal. Combining various omics datasets with computational approaches has helped uncover synthetic lethal relationships in cancer. For example, authors integrated genomic, transcriptomic, and proteomic data to predict synthetic lethal gene partners of the tumor suppressor gene TP53. The computational studies showed that if PARP is suppressed in TP53-mutant cancers, it results in synthetic lethality, and thus, PARP inhibitors like Olaparib are used in practice [167].

5.3. Targeting metabolic pathways in cancer

Cancer cells are known to have a unique metabolic phenotype in order to sustain their growth and division. Such metabolic reprogramming events have been elucidated by integrating the metabolomic and transcriptomic data sets. Cancer cells were analyzed by employing multi-omics approaches to delineate metabolic enzymes and pathways that are upregulated. These datasets were subjected to computational analysis for the purpose of identifying and ranking metabolic targets for drug development. For example, the IDH1 gene was considered as a potential therapeutic target in gliomas with IDH1 mutation. Specific IDH1 inhibitors like Ivosidenib (Tibsovo) have been designed and have proven effective in the treatment of IDH1-mutant cancers [168].

5.4. Comprehensive cancer genomic profiling

The TCGA project has obtained extensive multi-omics data of multiple cancers at the genomic, transcriptomic, proteomic, and epigenomic levels. These datasets have been combined and processed with the help of bioinformatics tools to identify new cancer drivers and targets for

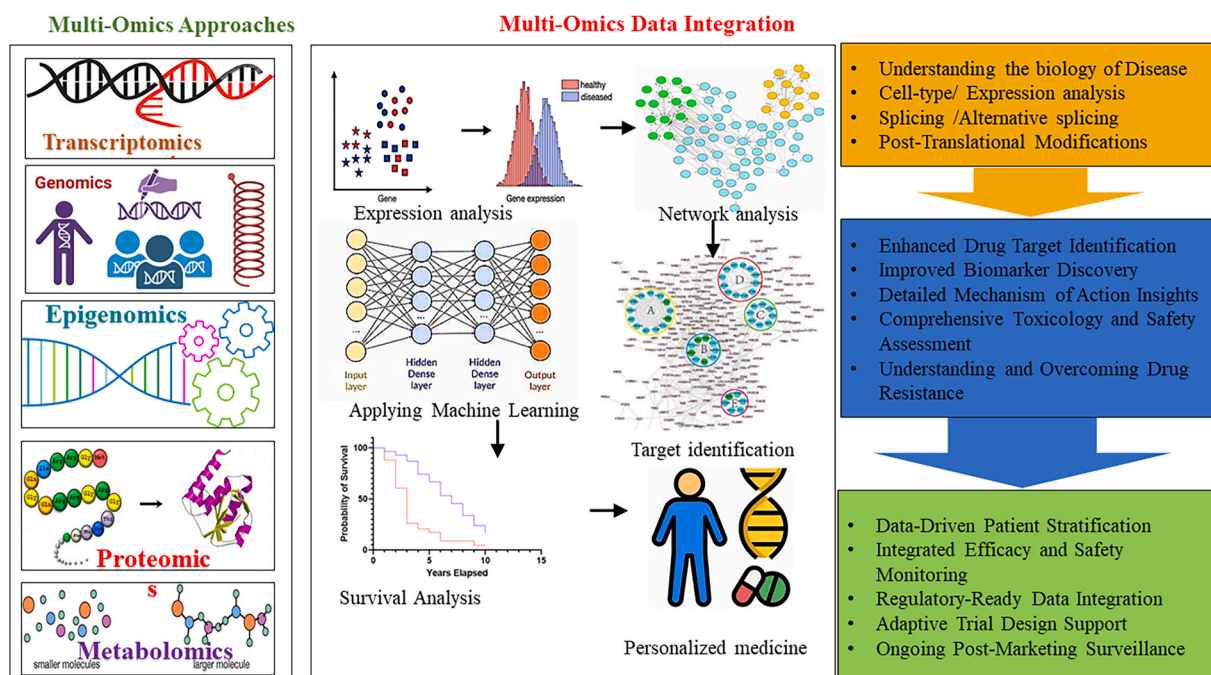


Fig. 5. Illustration of combined use of multi-omics data (genomics, proteomics, metabolomics) and computational tools (molecular docking, QSAR models, virtual screening, pharmacophore modeling, machine learning, AI, and molecular dynamics simulations) to enhance the development of targeted anticancer therapies.

treatment. For example, when performing the integrative analysis of TCGA data, it was identified that HER2 gene is overexpressed in certain subtypes of breast cancer and, therefore, the HER2-targeted therapy such as Trastuzumab (Herceptin) has been developed [169]. This introduction of personalized treatment has therefore helped the patients with HER2-positive breast cancer. Using multi-omics data along with computational solutions, the current process of seeking cancer drugs has dramatically changed due to a better perception of the disease. It is useful in discovering potential targets for new drugs, explaining the causes of the disease at the molecular level, and designing treatment plans. For instance, the identification of BRD4 inhibitors, synthetic lethality relationships, cancer metabolic pathways, and cancer genomics show the possibility of multi-omics in oncology [170]. These advancements in computational methods and multi-omics technologies will improve the search for better cancer treatments in the future.

6. Revolutionizing oncology: the impact of deep learning on anti-cancer drug discovery

Machine learning, with the use of many layered neural networks, has become instrumental in the process of searching for new anticancer drugs [171]. This capability of automatically learning and extracting feature from large and complex data set makes ANN ideal suited for the complex nature of new cancer therapeutic discovery. Deep learning analyze various types of data such as genomic sequences, molecular structures, high-throughput screening data, and clinical outcomes and may enhance the multiple steps of the drug discovery process [172]. Modern deep learning methods, including CNNs, RNNs, and autoencoders, have been used in the field of drug discovery starting from target identification and validation up to compound identification and optimization [173]. These models are capable of analyzing high-dimensional data, picking out intricate features, and predicting the biological activity, toxicity, and pharmacokinetics of lead compounds (Fig. 6). For example, in a CNN, the output of each layer is computed using the equation:

$$O_{i,j,k} = f\left(\sum_{m,n,l} \left(I_{i+m,j+n,l} \cdot K_{m,n,l,k}\right) + b_k\right) \quad (4)$$

where $O_{i,j,k}$ is the output at position (i,j) in the k -th feature map, I is the input, K is the kernel, b_k is the bias for the k -th feature map, and f is the activation function.

Similarly, in RNNs, the hidden state h_t at time t is computed as:

$$h_t = f(W_{hx}x_t + W_{hh}h_{t-1} + b_h) \quad (5)$$

where x_t is the input at time t , W_{hx} and W_{hh} are weight matrices, b_h is the bias, and f is the activation function.

Autoencoders, used for dimensionality reduction and feature extraction, consist of an encoder and a decoder, where the encoding process is represented as:

$$h = f(Wx + b) \quad (6)$$

and the decoding process as:

$$\hat{x} = f'(Wh + b') \quad (7)$$

where $\hat{x}$ is the input, h is the encoded representation, W and W' are weight matrices, b and b' are biases, and f and f' are activation functions. The other crucial use of deep learning in drug discovery is in estimating the binding affinity of small molecules to their biological targets. It basically entails feeding deep learning algorithms with large databases of known protein-ligand interactions to estimate the binding energy of novel molecules. The binding affinity (K_d) can be related to the binding free energy (ΔG_{bind}) using the equation:

$$\Delta G_{bind} = -RT \ln K_d \quad (8)$$

The deep learning algorithm learns to identify the patterns that define the binding affinity, including geometric aspects and molecular charges. The loss function L used to train these models to minimize prediction error is often expressed as: $L = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$, where y_i is the actual binding affinity, $\hat{y}_i$ is the predicted binding affinity, and n is the number of training examples.

6.1. Prediction of drug-target interactions

The prediction of drug-target interactions (DTIs) is an important step in the drug discovery process since it aims to identify potential

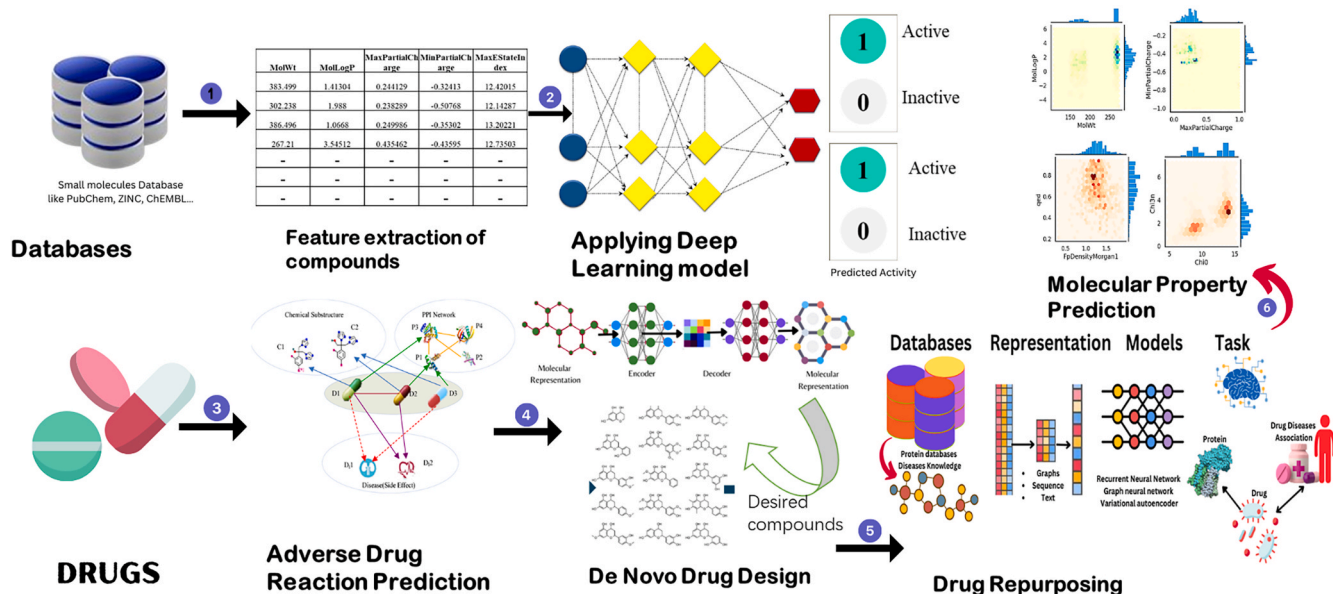


Fig. 6. Integration of informatics, deep learning, and pharmacological testing in the drug development process. It starts with a database of small molecules from sources like PubChem, ZINC, and ChEMBL. Key molecular properties such as molecular weight, logP, and electronic properties are extracted as features. These features are then input into a deep learning model, represented by neural networks, to predict the activity of these compounds. The model's output categorizes compounds as either active or inactive.

therapeutic compounds that may interact with certain biological targets like proteins in order to induce a preferential biological effect [174]. Current approaches to estimate DTIs include high-throughput screening and biochemical assays, which are cumbersome and costly. On the other hand, deep learning presents a computational and financially effective way to predict DTIs based on big data and patterns of successful encounters [175]. Several deep learning models such as CNNs, RNNs, and CNN-RNN hybrids are used to accurately predict the binding affinity and efficacy of the drug candidates. CNNs are particularly efficient in working with structured data such as images and sequences which are applicable in predicting DTIs due to the structural characteristics of drugs and targets. A notable example is DeepDTA model that uses CNNs to predict the binding affinity between a drug and its target [176]. DeepDTA takes as inputs SMILES strings of the drug compounds and amino acid sequences of the target proteins and is trained on pairs of them. From these representations, the CNN predicts binding affinity while surpassing traditional methods.

RNNs are particularly effective for sequential data and have been applied to predict DTIs based on sequences in molecular structures and protein sequences. For instance, the use of long short-term memory (LSTM) networks to capture dependencies within drug and protein sequences can help predict binding interactions better. Other approaches that use both CNNs and RNNs have also been found to be useful in DTI prediction. These models take advantage of CNNs for feature extraction and RNNs for sequence modeling. For instance, a hybrid model may employ CNNs in the spatial domain of the drug molecule, and RNNs in the temporal domain of a protein target to enhance the accuracy of the model [177].

6.2. Case studies (DeepDTA: drug-target binding affinity prediction)

As mentioned earlier, DeepDTA employs CNNs to determine the binding propensity of drugs to targets. The model was trained on the Davis and KIBA datasets which have experimentally measured binding affinities. The results obtained with DeepDTA signify the performance improvement of deep learning over traditional machine learning to predict the biological targets of molecular structures [176].

6.3. DeepBind: predicting protein-binding specificity

DeepBind is a deep learning-based tool to predict protein-binding specificity based on the DNA and RNA sequences. While DeepBind was designed to predict protein-nucleic acid interactions it can be generalized to drug-target interactions. DeepBind applies CNN on big data of binding sites, allowing the model to make predictions of binding affinity and specificity of proteins to different nucleic acid sequences [178].

6.4. AtomNet: structure-based drug design

AtomNet addresses structure-based drug design by using deep learning to predict small molecular binding to protein targets. AtomNet employs a CNN for the analysis of three-dimensional structures of molecules, which allows it to recognize suitable drug candidates reliably. It has been applied to discover new chemical entities for the targets like the Ebola virus and the MDM2-p53 protein-protein interaction and illustrates the usefulness of the method in drug discovery [179].

6.5. Generative models for drug design

Generative models are at the forefront of the drug discovery industry because they use deep learning to generate new drug like molecules. Some of these models can approximate the distribution of known drugs and create new compounds with certain properties. The most frequently employed generative models in drug design include GANs, VAEs, and RL-based models. These techniques provide a rich opportunity for

creating a new generation of drugs that are effective and have few side effects, thus dramatically shortening the drug discovery process [180].

6.6. Generative adversarial networks (GANs)

GANs consist of two neural networks: It consists of a generator and a discriminator. The generator produces new instances of data, while the discriminator compares them to real data. The two networks are trained simultaneously, where the role of the generator is to generate realistic data that are indistinguishable from the actual data with the discriminator aims at distinguishing actual data from the generated data. This adversarial process is useful to the generator in the sense that it can learn how to generate more realistic data. In other sectors of chemistry, GANs have been used in drug design to synthesize new molecules that possess certain characteristics. Kadurin et al. [181] presented druGAN, a GAN model for generating anticancer compounds. In druGAN, the generator network was trained using the database of known anticancer drugs, which makes it possible to create new molecules with high values of anticancer activity prediction. The model was successful in the synthesis of compounds that had similar chemical makeup as other drugs but had not been synthesized in the same way before.

6.7. Variational autoencoders (VAEs)

VAEs are also another type of generative model that is even more useful in drug design. A VAE consists of an encoder and decoder network. The encoder maps the input data into a lower dimensional representation of a latent space while the decoder maps the compressed data back into the input space of the original data. VAEs can reconstruct the input data and can also generate new samples from the latent space. When used for drug discovery VAEs can learn the distribution of known compounds and sample from this distribution to generate new molecules. Gómez-Bombarelli et al. [182] used VAEs for the generation of new molecules with certain chemical properties. According to the authors, to enable interpolation and synthesis of new compounds, molecular structures were embedded into a continuous representation space. This strategy resulted into the discovery of molecules with improved properties such as better binding and drug like attributes.

6.8. Reinforcement learning

Reinforcement learning-based models use RL concepts with generative models to improve molecular design. In this context, an agent (generative model) navigates through the chemical space and is awarded points depending on properties of the molecules created. The agent learns to generate compounds that bring in the highest reward, effectively finding new drugs with the best properties. A good example of RL in drug discovery is the work by Olivecrona et al. [], who used RL to generate new molecules with high activity against given targets. It was designed to be trained on a reward function which involves aspects such as binding affinity and drug like properties. This approach resulted in the identification of compounds with enhanced characteristics compared to compounds identified using conventional techniques.

In addition, Merk et al. [184] and David et al [185], employed deep generative model to develop polypharmacological agents that target multiple proteins. The model employed the GANs and VAEs to generate molecules with the desired multiple targeting properties. It made it possible to identify multi-targeting compounds, and considering this, it can be assumed that generative models can help design complex drugs for cancer treatment.

6.9. Predicting synergistic effect of drugs

The identification of synergistic drug pairs is very relevant in the development of combination therapies especially in cancers where several drugs neutralize the resistance factors and treat different

signaling pathways. Current approaches for the discovery of synergistic drug combinations primarily rely on cumbersome experimental assays. However, deep learning represents a much more effective solution, as it predicts synergy based on drug properties, biological activity, and genomic information (Table 4). Deep Synergy is an example of the deep learning approach that can be used to predict synergistic drug interactions. It involves features like drug-target binding affinities, gene expression data, and chemical signatures to estimate the drug-drug interactions. As DeepSynergy is trained with a large number of drug pairs, it can be used to find potential synergistic combinations thus minimize the need for experimental validation [186,187]. In a pioneering work, Preuer et al. [186] proposed DeepSynergy for the synergistic effects of drug pairs for cancer treatment. The model was trained on the data obtained from the National Cancer Institute's ALMANAC project, which includes the information about how thousands of drug combinations affect various cancer cell lines. DeepSynergy correctly selected new synergistic drug combinations which were then confirmed through lab experiments, which proves that the model can be helpful for cancer treatment combination therapy discovery. In the treatment of leukemia, Zitnik et al. [188] utilized deep learning to predict drug interactions. To model drug synergy, they used multi-omics data sets which include the transcriptome and proteome. The model also proposed several other drug combinations with high synergy index as revealed from the patient-derived leukemia cells. These predictions were confirmed experimentally and have determined the effective drug combinations which could be useful for the improvement of the leukemia treatment outcomes.

Batistta et al. [189] applied deep learning to forecast synergistic drug combinations for cancer. The model was developed using a large set of drug response profiles and genomic data for cancer cell lines. Combining deep learning with genomic characteristics made it possible for the model to predict multi-component combinations that affect genetic mutations common in cancer. A few predicted combinations were further tried in vitro, where significant synergistic effects were observed; suggesting new approach to the treatment of NSCLC cancer.

Recently, deep learning coupled with cheminformatics has been used in the identification of the synergistic anticancer agents. The authors used a deep learning model to predict chemical properties and biological activities of several compounds. Based on the model predictions, several

drug pairs showing synergistic effects were selected and evaluated in cell-based assays. Specifically, combining a kinase inhibitor with a DNA-damaging agent yielded a synergistic effect in killing cancer cells compared with monotherapy with either drug. Deep learning in drug synergy for combination therapy is transforming the process of identifying synergistic drugs in cancer treatment [190]. Such models as DeepSynergy showed that it is possible to combine different types of data and make accurate predictions regarding synergistic drug interactions. Leukemia and NSCLC provide real-life examples of how this strategy works and has advantages. PK/PD integration and other subsequent developments hold the potential to improve the prediction and validation of synergistic combinations of drugs in the future to achieve better therapeutic outcomes in cancer.

6.10. New perspectives for implementation of deep learning for anti-cancer drug discovery

Transfer learning entails using a model that has been trained for a particular task then fine-tuning it for a unique task. For DTI prediction, transfer learning might be applied to use models created on large and general datasets for enhancing the prediction on small and disease-specific datasets. This approach can improve the accuracy and speed of DTI predictions, which is especially valuable for rare diseases with scarce information available. Thus, the investigation of deep learning model's decision-making process could help to shed light on the biological processes of drug-target interaction. Interpretability methods for deep learning models attempt to identify what features of the input data most influenced the model's predictions. In turn, this transparency enables researchers to check the model's predictions and inform further experimental design.

Active learning where the model is trained in cycles, during which the most informative data points for labeling and adding into the training set are chosen. In DTI prediction, active learning may assist in ranking the drug candidates which the model is least confident about. This approach can help to maximize the use of experimental resources and minimize the time needed to discover a new drug. Deep learning has emerged as an effective method for the prediction of drug-target interactions by analyzing big data. CNNs, RNNs, and their combination have been proved to be more effective in predicting binding affinities

Table 4

Tools/Strategies used deep learning for drug discovery.

Tool/Method	Type	Primary Application	Key Features	References
DeepDTA	Convolutional Neural Network (CNN)	Predicting Drug-Target Interactions	Uses SMILES strings and protein sequences for binding affinity prediction	[189]
DeepBind	Convolutional Neural Network (CNN)	Predicting Protein-Binding Specificity	Analyzes DNA/RNA sequences to predict binding affinity and specificity	[191]
AtomNet	Convolutional Neural Network (CNN)	Structure-Based Drug Design	Processes 3D molecular structures to predict bioactivity	[179]
drUGAN	Generative Adversarial Network (GAN)	Generative Drug Design	Generates novel anticancer compounds	[181]
DeepSynergy	Hybrid (Deep Learning + Data Integration)	Predicting Drug Synergy	Integrates drug-target interactions, gene expression profiles, and chemical structures	[186]
DeepTox	Recurrent Neural Network (RNN)	Predicting Toxicity	Predicts toxicological profiles from chemical structures	[192]
Chemprop	Graph Neural Network (GNN)	Molecular Property Prediction	Predicts molecular properties from graph representations of molecules	[193]
MoleculeNet	Benchmark Dataset + Various Models	Evaluating Molecular Machine Learning Methods	Provides benchmark datasets and models for evaluating ML methods in drug discovery	[194]
DeepChem	Framework + Various Models	Drug Discovery	Provides deep learning tools and models for cheminformatics and bioinformatics	[195]
REINVENT	Reinforcement Learning (RL)	Molecular Design	Uses RL to optimize de novo molecular design	[196]
Graph2Seq	Graph Neural Network (GNN)	Molecular Sequence Prediction	Converts molecular graphs to sequences for downstream tasks	[]
BioAutoML	Automated Machine Learning (AutoML)	Automated Model Selection and Optimization	Uses AutoML for model selection and hyperparameter tuning in bioinformatics	[198]
MT-DTI	Multitask Deep Learning	Predicting Drug-Target Interactions	Uses multitask learning to predict interactions across multiple targets	[199]
Deep GraphMolGen	Graph Convolutional Network (GCN)	Generative Molecular Design	Generates molecules with specific properties using GCNs	[200]
Synergy Finder	Platform + Deep Learning Models	Drug Synergy Analysis	Provides tools and models for analyzing drug synergy	[201]

and exploring potential drugs. DeepDTA, DeepBind, and AtomNet are some examples of how deep learning can revolutionize this area. Directions for the future such as transfer learning, explainable AI, integration with multi-omics data, and active learning would lead to even more efficient and accurate predictions of DTI which in turn would mean better cancer treatment options.

7. Unveiling anti-cancer properties of compounds with computational tools

Metabolites have been considered a fertile ground for the discovery of anticancer agents because of their structural and functional diversities. These bioactive compounds include those from plants, marine organisms, and microorganisms where they present multiple actions against cancer through apoptosis, anti-proliferation, and regulation of cancer-related signal transduction [202]. Several natural products have been shown to possess anticancer activity and have been used as drugs (Fig. 7). For example, Paclitaxel (Taxol) from the Pacific yew tree binds with microtubules and inhibits cell division, which is useful in treating breast, ovarian, and lung cancers [203]. Likewise, Camptothecin, derived from the Chinese tree *Camptotheca acuminata* has DNA topoisomerase I and causes DNA injury and cell demise [204]. Irinotecan and Topotecan are some of the products derived from it which are used in the treatment of colorectal and ovarian cancer [205,206]. Other examples of natural compounds that have been found to possess anticancer properties are curcumin derived from turmeric and Epigallocatechin Gallate (EGCG) from green tea showing that natural products are of great potential in cancer treatment.

Software applications have emerged as vital aids in the identification and synthesis of anticancer compounds from natural products. These tools help in some stages of drug discoveries such as virtual screening, molecular docking, pharmacophore modeling, and QSAR. Thus, the application of computational methods allows the researchers to exclude the time and money-consuming experimental approaches and identify and optimize the natural compounds with the potential anticancer activity. The recent integration of databases and computational tools has significantly accelerated the pace of drug discovery in cancer therapeutics. Databases like NPACT and TCMSP provide invaluable resources

on natural and traditional compounds, their mechanisms, and therapeutic potentials against cancer, enabling researchers to screen and identify promising candidates more effectively. SuperNatural II and CPCD offer extensive libraries of natural compounds and phytochemicals with documented anti-cancer properties, further supporting targeted drug development efforts. Moreover, software tools like AutoDock Vina and CB-Dock facilitate precise molecular docking studies, crucial for understanding drug-receptor interactions and optimizing drug efficacy. The application of QSAR modeling tools and PharmMapper allows for the prediction of biological activity and potential drug targets based on molecular structures, enhancing the selectivity and effectiveness of new drug candidates (Table 5). Collectively, these tools exemplify the powerful synergy between computational resources and traditional pharmacological knowledge, driving forward the discovery of novel cancer therapies with higher precision and efficiency.

8. Perturbation-theory machine learning (PTML) in multi-omics for cancer drug discovery

8.1. The role of PTML models in cancer drug discovery

Machine learning has contributed significantly to advancing cancer drug discovery through the integration of multi-omics data. One of the most groundbreaking approaches in this context is perturbation-theory machine learning (PTML) [235,236]. PTML models have demonstrated immense potential in predicting a wide array of biological outcomes by integrating diverse data types [237]. These models have revolutionized traditional QSAR (quantitative structure-activity relationship) approaches by introducing multi-target QSAR, multi-condition QSAR, and multi-tasking quantitative structure-biological effect relationship (mtk-QSBER) models, enabling more comprehensive drug screening strategies [238].

8.2. Multi-target and multi-condition QSAR models

Traditional QSAR models primarily focus on predicting the activity of compounds against a single target or biological endpoint. In contrast, multi-target QSAR models within the PTML framework enable the

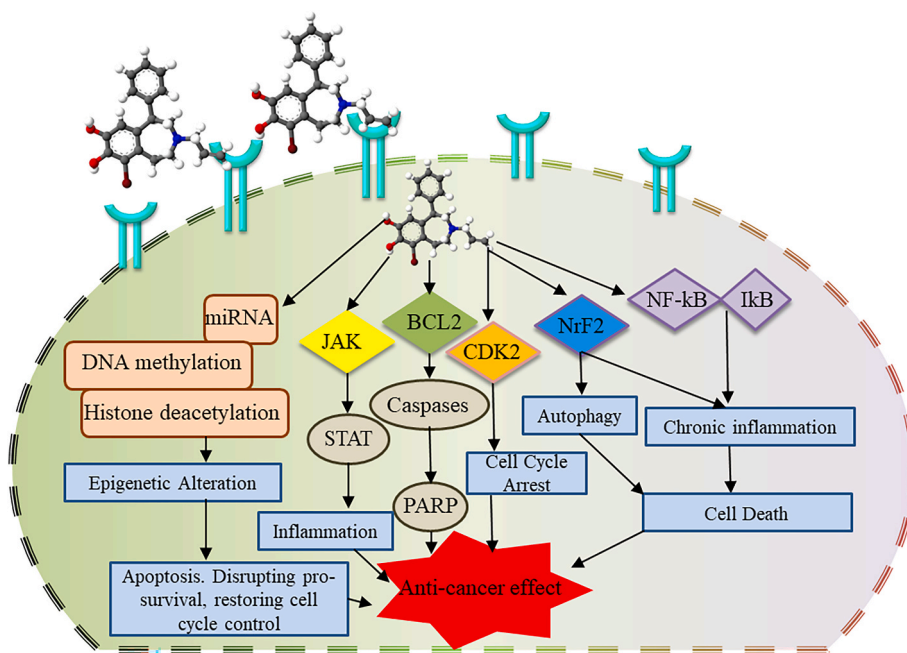


Fig. 7. Illustrates the intricate molecular pathways through which phytochemicals exert anti-cancer effects, mediated by epigenetic and signaling mechanisms. The diagram showcases a variety of herbs and their extracted phytochemicals, leading to multiple biological impacts. These compounds interact with key molecular targets such as microRNAs, BCL2, JAK/STAT, and NF-κB pathways.

Table 5

Comprehensive list of web tools, software, and databases for compounds used for cancer treatment.

Tool/Database	Type	Primary Application	Key Features	References/ Developers
Database				
NPACT	Database	Natural plant compounds in cancer	Contains information on natural compounds with anticancer activity	[207]
TCMSP	Database	Traditional Chinese Medicine compounds	Provides information on pharmacokinetics and molecular mechanisms	[208]
SuperNatural II	Database	Natural compounds	Contains structures and activities of natural products	[209]
IMPAT	Database	Phytochemical information	Database of phytochemicals with biological activities	[210]
NPASS	Database	Natural product activity and species source	Information on bioactivity and source of natural products	[211]
ChemSpider	Database	Chemical structure database	Provides access to over 58 million structures, properties, and biological activities	[212]
PubChem	Database	Chemical information	Open chemistry database with compound structures and bioactivities	[213]
ChEMBL	Database	Bioactive molecules	Database of bioactive molecules with drug-like properties	[214]
NCI Database	Database	Cancer-related compounds	Compounds screened for anticancer activity by the National Cancer Institute	[215]
CancerDR	Database	Cancer drugs and natural compounds	Contains drug sensitivity data for various cancer drugs and natural compounds	[216]
ZINC	Database	Chemical compounds	Free database of commercially-available compounds for virtual screening	[217]
BindingDB	Database	Binding affinity data	Database of measured binding affinities for drug-like molecules and protein targets	[218]
STITCH	Database	Chemical-protein interactions	Provides interactions between chemicals and proteins	[219]
Herbal Ingredients' Targets Database (HIT)	Database	Herbal ingredients	Database of targets for herbal ingredients	[220]
Traditional Chinese Medicine Integrated Database (TCMID)	Database	TCM compounds and targets	Database of traditional Chinese medicine compounds and their targets	[221]
Software				
Toxicity Estimation Software Tool (TEST)	Software	Predicting toxicity	Estimates the toxicity of chemical compounds based on structure-activity relationships	[222]
PASS Online	Software	Predicting biological activity	Predicts the biological activity spectrum of compounds	[223]
QSAR Toolbox	Software	QSAR modeling	Assists in the development and application of structure-activity relationships	[224]
Molinspiration	Software	Molecular property calculation	Provides tools for calculating molecular properties and bioactivity scores	[225]
AutoDock Vina	Software	Molecular docking	Open-source program for performing docking simulations	[76]
PyMOL	Software	Molecular visualization	Open-source molecular visualization system for rendering and animating molecules	[82]
ChemBioDraw	Software	Chemical drawing and analysis	Tool for drawing chemical structures and predicting molecular properties	[226]
LigandScout	Software	Pharmacophore modeling	3D pharmacophore modeling software for drug discovery	[227]
Discovery Studio	Software	Drug discovery and design	Comprehensive suite for drug discovery, including docking, pharmacophore modeling, and QSAR	[103]
iGemDock	Software	Molecular docking	Integrated virtual screening and molecular docking platform	[103]
Open Babel	Software	Chemical toolbox	Converts chemical file formats and performs various cheminformatics tasks	[228]
RDKit	Software	Cheminformatics	Open-source cheminformatics software for drug discovery	[229]
WEB Tools				
SwissADME	Web Tool	ADME prediction	Predicts pharmacokinetics, drug-likeness, and medicinal chemistry friendliness	[102]
CB-Dock	Web Tool	Molecular docking	User-friendly docking service for predicting protein-ligand interactions	[230]
PharmaGist	Web Tool	Pharmacophore detection	Web server for detecting pharmacophores in ligand sets	[231]
NPDock	Web Tool	Protein-ligand docking	Tool for docking natural products with proteins	[232]
SIMCOMP	Web Tool	Chemical structure comparison	Tool for comparing chemical structures	[233]
PharmMapper	Web Tool	Reverse pharmacophore mapping	Tool for identifying potential targets for small molecules	[234]

simultaneous prediction of biological activities across multiple targets, such as different proteins, cell lines, or even organisms. Similarly, multi-condition QSAR models allow predictions to account for varying experimental conditions, such as different assay protocols. These models provide a more holistic view of a compound's potential efficacy and safety, enhancing the drug development process by enabling broader virtual screening applications [239,240].

8.3. Multi-tasking QSBER models

The multi-tasking quantitative structure-biological effect relationship (mtk-QSBER) models further enhance the PTML framework by simultaneously predicting multiple biological effects, including therapeutic activity, toxicological outcomes, and pharmacokinetic properties. This is especially valuable in cancer drug discovery, where a drug's efficacy must be balanced with its safety and tolerability [241,242]. By

integrating multiple biological tasks into a single model, PTML-based QSBER models streamline the evaluation of new compounds, reducing the time and cost required to bring a drug candidate from initial discovery to clinical trials.

8.4. Interpretability and molecular design

One of the critical advantages of PTML models is their interpretability. These models provide insights into the physicochemical and structural features that contribute to a compound's predicted biological effects. This interpretability is crucial for rational drug design, as it allows researchers to modify molecular structures in a targeted manner to achieve desired therapeutic outcomes [243–245]. For example, by understanding which molecular features enhance binding affinity to a specific cancer-related protein while minimizing toxicity, novel drug candidates can be designed more efficiently [246].

8.5. Applications of PTML in multi-omics

PTML models have broad applications in various subfields of multi-omics, offering significant advancements in cancer drug discovery. One key application is in chemoproteomics, where PTML models are used to study protein-small molecule interactions. This is crucial for target-based drug discovery, as these models help identify potential drug-protein binding sites, facilitating the discovery of inhibitors or activators that could be used in cancer therapy. Another important application of PTML models is in toxicogenomics, a field that analyzes gene and protein activity in response to toxic substances. By predicting how potential drugs might cause toxic effects at the genetic or proteomic level, PTML models help screen out candidates with undesirable side effects early in the drug discovery process, thus improving safety profiles.

In addition to these fields, PTML models are applied in cell-based omics, where the focus is on understanding how small molecules affect cellular pathways and processes. This is particularly valuable in cancer drug discovery, as it allows for the prediction of how a drug candidate might interact with cancer cells versus normal cells, providing insights into its potential efficacy and selectivity. Furthermore, the ability of PTML models to simultaneously predict multiple biological activities and conditions makes them powerful tools for virtual screening in cancer drug discovery. These models enable researchers to efficiently screen large compound libraries, identifying molecules with the desired therapeutic and safety profiles across multiple biological targets. By significantly reducing the need for extensive experimental validation, PTML models accelerate the drug discovery pipeline, making the process more efficient and cost-effective.

9. Future trends and conclusion

Deep learning, machine learning, and their application in clinical trials are promising developments for the future of anticancer drug discovery. They have the potential to improve the accuracy, speed and effectiveness of identifying and bringing new anticancer agents to the market. The use of deep learning, which is capable of training large and complex datasets and analyzing complex structures, will not cease to transform the drug discovery process in the near future. Future advancements are expected to thus address issues concerning interpretability of the deep learning models to enable the researchers to understand the predictions made by the models. This could include the creation of advanced architecture, for example, the attention mechanism, which helps to focus on the most significant features affecting model's predictions. Furthermore, connecting genomics, proteomics, transcriptomics, and metabolomics into deep learning models will offer a systemic understanding of cancer to find more unique drug targets and biomarkers for precise treatment.

Machine learning is going to become the crucial tool for predictive modeling and optimization in the drug discovery process. Current

mathematical models will be improved to predict the ADME topline and side effects of new drugs and their toxicity. One of the subfields in machine learning, active learning will gain more attention, whereby models can choose informative samples to train on. This will increase resource efficiency in experimental validation by directing activity to those molecules with the highest likelihood of success. In addition, machine learning algorithms will be used to develop better molecules with desired characteristics by anticipating the effects of changes in chemical structure. Integrating deep learning and machine learning with clinical trials is another significant future direction. It is possible to use these technologies for defining patients that are most likely to benefit from certain treatments, thus increasing the efficiency of clinical trials. By utilizing machine learning models, clinical trial data can be evaluated in real-time and facilitate changes to trial protocols on the fly. Known as adaptive clinical trials, this approach makes it easier and faster to develop new treatments by doing away with some of the most time-consuming processes of drug testing. This ongoing monitoring will help in ensuring that new therapies remain safe and effective once they are adopted in the ordinary populace.

Most of the innovations in deep learning and machine learning will be fostered by collaborative platforms and open-source tools in drug discovery. These platforms will enable the exchange of data, models, and experience between scientists across the world, thus encouraging more rapid advances. Frameworks – TensorFlow, PyTorch, DeepChem and others – will remain open-source to empower researchers with the necessary tools to advance and implement new models. Ethical and regulatory challenges will arise as AI and machine learning become increasingly involved in the discovery of new drugs and in clinical trials. The openness, impartiality, and responsibility of AI models will become critical factors in obtaining regulatory permission and public acceptance. Academics and stakeholders will need to come together and establish protocols and benchmarks for the appropriate integration of such technologies within the field of health care. Further, PTML models are expected to play an even more integral role in multi-omics and cancer drug discovery by integrating larger and more diverse datasets, such as real-world patient data and longitudinal studies, to further improve prediction accuracy. With advancements in deep learning and AI, PTML models will likely evolve to offer even greater interpretability, allowing for more precise drug design. Additionally, PTML models could be increasingly applied to personalized medicine, enabling the development of drugs tailored to individual genetic and molecular profiles. Their ability to predict multi-target interactions will also help uncover novel therapeutic pathways and drug combinations, ultimately leading to more effective and safer cancer treatments. To sum up, one can assert that the further development of deep learning, machine learning, as well as other computational tools, will determine the further course of anticancer drug discovery. These technologies will further our knowledge of cancer, aid in creating the next generation of cancer treatments, and make clinical research more efficient. Through adopting these advancements, the scientific community can enhance the discovery of effective treatments for cancer patients depending on their individual needs.

CRediT authorship contribution statement

Israr Fatima: Writing – original draft, Conceptualization. **Abdur Rehman:** Writing – review & editing, Writing – original draft. **Yanheng Ding:** Visualization, Software. **Peng wang:** Writing – review & editing, Data curation. **Yuxuan Meng:** Writing – review & editing, Data curation. **Hafeez Ur Rehman:** Writing – review & editing. **Dawood Ahmad Warraich:** Validation, Software. **Zhibo Wang:** Writing – review & editing. **Lijun Feng:** Writing – review & editing, Visualization, Validation. **Mingzhi Liao:** Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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The authors regret that the initially published graphical abstract does not fully reflect the key findings of our work. The authors have updated the graphical abstract to provide a more accurate visual summary and

respectfully request that the journal replace the current version with the attached revised graphical abstract.

The authors would like to apologise for any inconvenience caused.

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